



Dr Ngoc Tan Nguyen and his colleagues have created a battery that is both flexible and washable. It works even when twisted or stretched to twice its normal length, or after being washed multiple times (Credit: <https://news.ubc.ca>)

washable. It works even when twisted or stretched to twice its normal length, or after being tossed in the laundry. The battery comprises several ultra-thin layers of zinc and manganese dioxide fragments embedded in a polymer wrapped inside a casing of the same polymer. This construction creates an airtight, waterproof seal that ensures the integrity of the battery through repeated use. The battery can be integrated in devices worn next to the skin as it has a safer chemistry than lithium-ion batteries, which can produce toxic compounds when they break.

Safer radiation therapy with curved X-ray detectors

Researchers from the Advanced Technology Institute at the University of Surrey, in collaboration with Italy's University



Surrey researchers have identified key design rules for making curved X-ray detectors (Credit: www.surrey.ac.uk/news)

of Bologna, the National Physical Laboratory and Sheffield University, have identified design rules for a special class of 'inorganic in organic' semiconductors that can help in the creation of robust, curved digital detectors with high sensitivity, or digital film. The material has exceptional mechanical robustness and enables bending radii as small as 1.3mm. The development has huge potential in medical applications and can bring clearer and safer X-rays a step closer to reality.

Wireless device that detects soil health in 90 seconds

IIT Kanpur has developed a portable soil testing device called Bhu Parikshak that is based on near-infrared spectroscopy to detect real-time soil health in just 90 seconds through an embedded mobile application. The rapid soil



Soil testing device (Credit: www.iitk.ac.in)

testing device will assist individual farmers in obtaining soil health parameters of agricultural fields with recommended doses of fertilisers. The wireless soil testing device requires only 5 grams of dry soil sample for detecting macronutrients and connects itself with the mobile through Bluetooth.

First of its kind hydrogen-powered e-bike

Dutch design office Studio MOM has collaborated with the Australian hydrogen energy storage solutions provider LAVO to develop an e-bike that will be hydrogen-powered and carry heavy cargo over long distances without requiring heavy batteries. Called the LAVO bike, the vehicle leverages the higher energy-to-weight ratio offered by hydrogen fuel cells in contrast with lithium batteries typically used